

MAJOR CONTRIBUTORS TO THIS REPORT



Madhukar Tripathi
regional manager,
Anritsu India Pvt Ltd



Manish Joshi
deputy CEO, Sciencetech
Technologies Pvt Ltd



Satish Mohanram
technical marketing
manager, National
Instruments India



Stephen Hire
general manager-India,
Aeroflex



T. Anand
principal consultant,
Knewron



Vishal Gupta
senior application consultant
(RF/MW, Surveillance),
Agilent Technologies

moving towards an approach whereby there are people who are developing platforms and people who are developing apps for the platforms. Also, programmable electronics is entering every part of life today, which is driving the need for being able to do other testing and really high-speed testing. "Those are two factors which are driving the growth of signal generators and I think that FPGA signal generators are an outcome of the need that we see in the market," says Mohanram.

On a similar note, Vishal Gupta, senior application consultant (RF/MW, Surveillance), Agilent Technologies, informs that the driving factors are emerging commercial communication technologies, enhanced defence requirements and recreation of real-time field signals in the lab, and the restraining factors are hardware limitations, faster DAC and high component costs.

In the future

With so many different technologies that are being worked on and their rapid evolution, flexible signal gen-

erators need to be developed to ensure their long and useful working life. Developing common hardware platforms and software-defined radio techniques enable the same hardware to address many applications with customisable software that can be upgraded as technologies change. Modular platforms and software-defined radios also help companies protect their long-term investment.

The FPGA capability on the signal generator has provided the designer with a lot of new capabilities to do different types of testing. Mohanram says, "For instance, the concept of hardware-in-loop testing that people have been trying to incorporate to address the needs of today's electronic equipment, is being achieved really well by these signal generators that have been enabled by FPGAs that are playing a significant role."

Signal generators phased from controls and knobs on the front panel to software, but today people are able to develop custom test units which get connected to say a gyroscope. One

can even go to the extent of simulating the whole gyroscope on the FPGA and generating the signal as though it comes from the gyroscope from the signal generator. IPs are being developed which can simulate all of these different functionalities and these are also being shared across different industries. "This I think is going to be the future where you will see a whole bunch of these IPs available for download and tweaked on these signal generators. Thereby making these signal generators behave as though they are a specific sensor or an instrument," adds Mohanram.

Talking about touch screens in signal generators, Hire says, "Smartphones have led the world in showing how touch screens can simplify and improve the user experience. And we believe that touch screen control of test equipment also has the potential for delivering big productivity gains in R&D and production." ●

The author is a senior technical correspondent at EBY, Bengaluru

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou



Your favourite Magazine on Open
Source is now on the Web, too.

LinuxForU.com

Follow us on Twitter@LinuxForU